In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE (Advanced), 2017

ALL INDIA TEST SERIES

FULL TEST – V

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 10)** contains 10 multiple choice questions which have one or more than one correct answer. Each question carries **+3 marks** for correct answer. There is no negative marking.

(ii) **Section-C (01 – 10)** contains 10 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

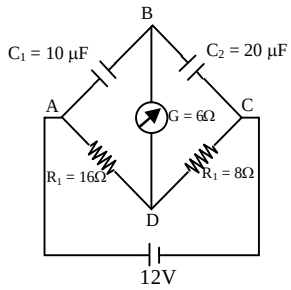
SECTION – A

Straight Objective Type

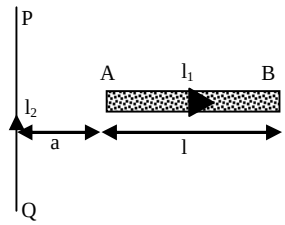
This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

1. Four point charges +q, +q, – q, and – q are placed respectively at corners A, B, C and D of a square. Then
   (A) the potential at the centre O of the square is zero
   (B) the electric field at the centre O of the square is zero
   (C) If E is the mid-point of side BC, the work done in carrying an electron from O to E is zero
   (D) If F is the mid-point of side CD, the work done is carrying an electron from O to F is zero

2. In the circuit show in figure, if the galvanometer resistance is 6 Ω, then in the steady state
   (A) no current flows through the galvanometer
   (B) the current through R_2 is 4A
   (C) the charge on C_1 is 80 μC.
   (D) the charge on C_2 is 80 μC

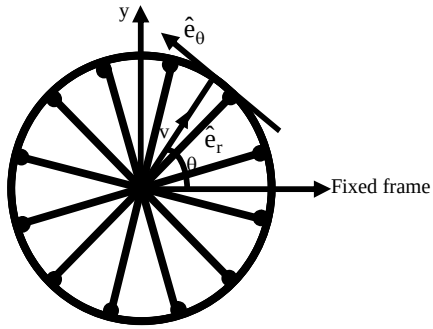


3. A thin rod AB of length l carries a current I_1. It is placed in the magnetic field of a long wire PQ carrying a current I_2 as shown in figure
   (A) The force experienced by the rod is $F = \frac{\mu_0 I_1 I_2}{4\pi} \log_e \left(1 + \frac{l}{2a}\right)$
   (B) The force experienced by the rod is $F = \frac{\mu_0 I_1 I_2}{4\pi} \log_e \left(1 + \frac{l}{a}\right)$
   (C) The rod experiences no torque
   (D) The rod experiences a force as well as a torque.



Space for rough work

4. A fly crawls with constant speed v along the radial spoke of a wheel which is rotating with constant angular speed ω. Initially the fly starts from centre of the wheel along x-axis. Then (where $r = v\theta$)



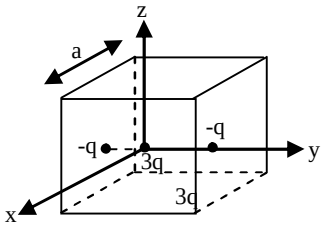
(A) The actual velocity of the fly in fixed frame is $\vec{v} = v\hat{e}_r + r\omega\hat{e}_\theta$

(B) The actual velocity of the fly in fixed frame is $\vec{v} = \sqrt{v\theta\omega}\hat{e}_r + r\omega\hat{e}_\theta$

(C) The position of fly at any instant $\vec{r} = \frac{v.\theta}{\omega}\hat{e}_r$

(D) The position of fly at any instant $\vec{r} = \frac{vt}{\omega}\hat{e}_r$

5. A cubical region of side a has its centre at the origin. It encloses three fixed point charges, -q at (0, -a/4, 0) + 3q at (0, 0, 0) and -q at (0 + a/4, 0) Choose the correct options(s)



(A) The net electric flux crossing the plane x = +a/2 is equal to the net electric flux crossing the plane x =-a/2

(B) The net electric flux crossing the plane y = +a/2 is more than the net electric flux crossing the plane y =- a/2

(C) The net electric flux crossing the entire region is q/ε_0

(D) The net electric flux crossing the plane z = +a/2 is equal to the net electric flux crossing the plane x = +a/2

6. Let the energy, magnitude of linear momentum and angular frequency of an electron in hydrogen atom be E,P and ω respectively. If n be the corresponding quantum number, then

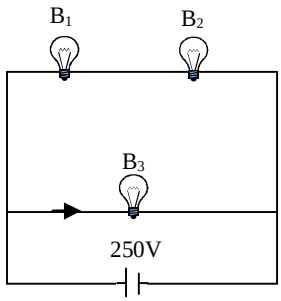
(A) E/W varies as n
(B) E P/W is independent of n
(C) P W varies as $n^{1/2}$
(D) E P W is independent of n.

Space for rough work

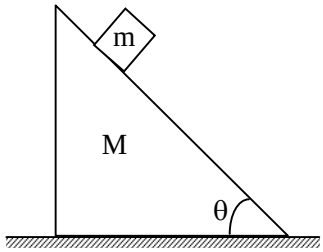
7. A nitrogen nucleus $^{14}N_7$ absorbs a neutron and can be transformed to lithium nucleus $^{6}Li_3$ under suitable condition, after emitting
(A) 4 proton and 3 neutrons
(B) 5 proton and 1 beta particle
(C) 2 alpha particles, 1 neutron and 2 gama photons
(D) 4 protons and 4 neutrons

8. At 0^oC a black body placed in 0^oC surrounding then it emits
(A) all wavelengths of radiation (B) only visible light
(C) only micro wave radiation (D) no radiation

9. A 100 W bulb B_1, and two 60 W bulbs B_2 and B_3, are connected to a 250 V source, as shown in figure. Now W_1, W_2 and W_3 are the output powers of the bulbs B_1, B_2 and B_3, respectively. Then
(A) $W_1 > W_2 = W_3$ (B) $W_1 > W_2 > W_3$
(C) $W_1 < W_2 = W_3$ (D) $W_1 < W_2 < W_3$



10. A block of mass m is placed on a prims of mass M. The inclined surface is smooth and inclination with horizontal is θ. The horizontal surface is sufficiently rough to prevent slipping of prism. The body of mass *m* is coming down the inclined face then
(A) Acceleration of body along the inclined surface is g sin θ
(B) Frictional force is $\frac{1}{2}mg\sin 2\theta$
(C) Maximum frictional force is $\frac{mg}{2}$
(D) Friction force will be maximum when $\theta = 45^0$



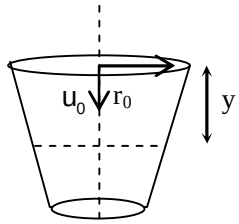
Space for rough work

SECTION – C

(One Integer Value Correct Type)

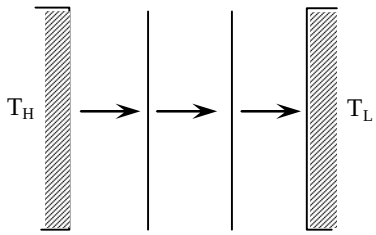
This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Water flows vertically down ward out of a round nozzle of radius r_0 with a speed u_0. Which is not quite large for some distance below nozzle the flow in water laminar stream over this distance the stream radius r_y as a function of the distance of belong the nozzle is $r_y = r_0\left[1+\eta y\right]^{-1/\alpha}$. Value of $(\eta+\alpha)$ is $\left(\text{take value of } u_0^2/g = 1\right)$.



2. A 60 cm metal rod M_1 is joined to another 100 cm metal rod M_2 to form an L shaped single piece This piece is hung on a ply at the joint. The two rods are observed to be equality inclined to the vertical. The two rods are equally thick. If the ratio of density M_1 to that of M_2 is η, then value of $\frac{18\eta}{25}$ is

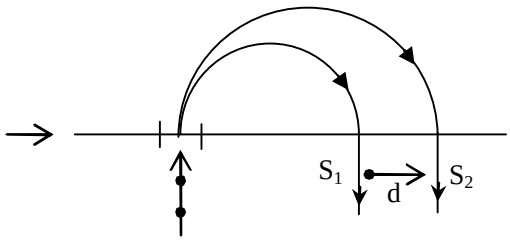
3. A black surface at constant high temperature T_H is parallel to another black plane surface at a constant lower temperature T_L (enclose vacuum in between). A heat shield consisting of two thin black plates is placed between the warm and the cold surfaces and parallel to these. After some time steady state is achieved. The value of $\eta = \frac{J_0}{J}$ is (J_o is initial heat current and J is final current)



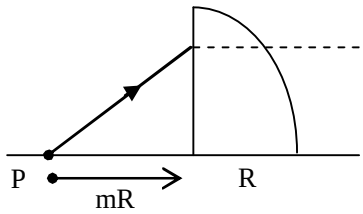
Space for rough work

4. The velocity filter of a mass spectrometer uses a uniform E = 100 v/m and a magnetic field $B_1 = 2\times10^{-2}$ T below filter at the uniform magnetic filed of induction 8×10^{-2} used to deflect the beam above filter. Both the beams are +e charged having mass numbers 20 and 22 pass through the filter and make a 180° turn in the deflecting filed. If the distance between the points S_1 & S_2 is d, then find the approximate (integer) value of 2d?

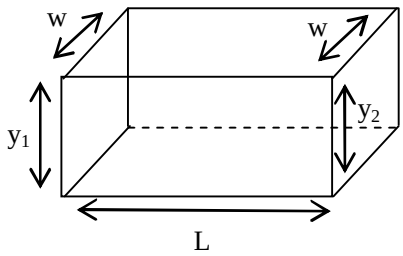
(Value of $\frac{1amu}{e} = 10^{-8}$ SI unit) (Note: spectrometer does not changes speed of ions)



5. A quarter cylinder of radius R and refractive index 1.5 is placed on a table as shown in figure. For what value of 3m, a ray from P will emerge parallel to table? Assume small aperture.

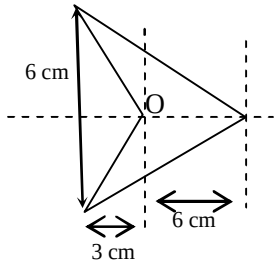


6. A material with uniform restively ρ is formed in the shape of a wedge as shown. The resistance of wedge across length L is $R = \frac{\rho\eta}{L}\ln\eta'$. Then find the value of $\frac{\eta}{6}+\eta'$. $\left(y = \frac{L}{3}, y_2 = \frac{L}{6}, W = \frac{L}{6}\right)$

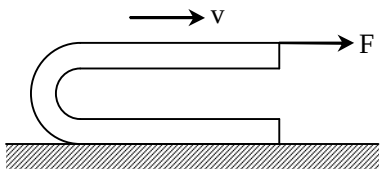


Space for rough work

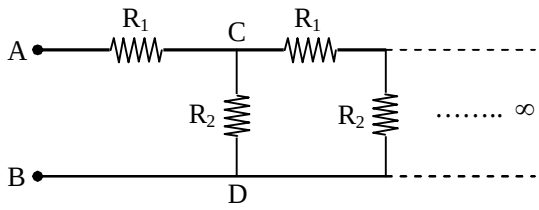
7. A uniformly thick plate in the shape of an arrowhead has dimension as shown. The centre of mass lies at a distance x cm from O. The value of x is



8. A long flexible inextensible rope of uniform mass density λ is being pulled on a rough floor with horizontal force F in such a way that the lower part is at rest and upper part moves with constant speed v as shown, the magnitude of F will be (in N) . $(\lambda = 1\,kg/m, v = 2m/s)$



9. Consider an infinite ladder net work as shown in figure A voltage is applied between points A and B. If the voltage is hawed after each section, find the ratio R_2/R_1.



10. Value of electric field at centre of charged hemisphere is ……. N/C if it has uniform charged density $\delta = 8\varepsilon_o c/m^2$ at surface.

Space for rough work

Chemistry

PART – II

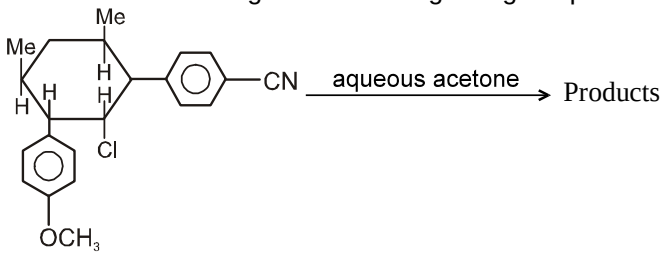
SECTION – A

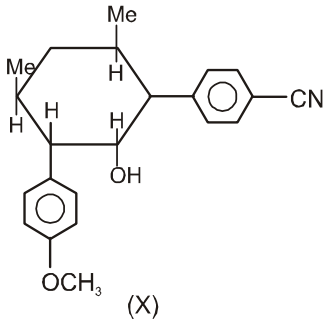
Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

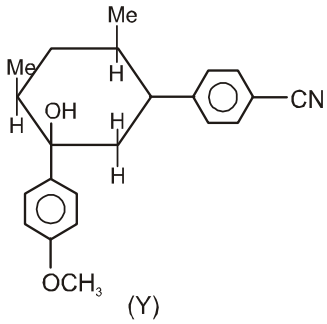
1. A system is changed from an initial state to a final state by a manner such that if the same change from the initial state to the final state were made by a different path, which of the following statements are correct for ΔH system?
(A) remain the same
(B) will depend upon the type of path
(C) Heat exchanged q will be same if the path is isobaric
(D) Heat exchanged q will be different if the path is not isobaric

2. Which of the following are correct regarding the products of the given reaction?

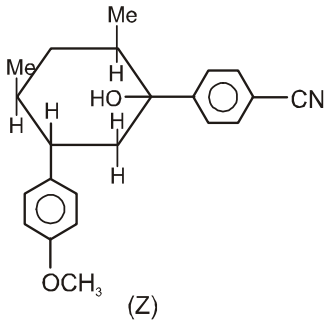




(X)



(Y)



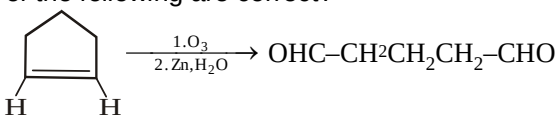
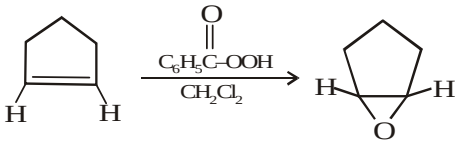
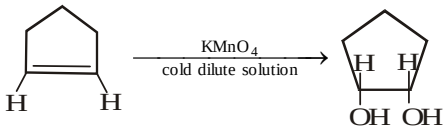
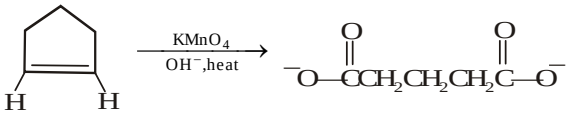
(Z)

(A) only X and Z are formed
(B) Y > Z (amount)
(C) X > Y (amount)
(D) Y > X (amount)

Space for rough work

3. Which of the following pairs of nitrates give the same gaseous products on thermal decomposition?
(A) KNO_3 and $Pb(NO_3)_2$ (B) KNO_3 and $NaNO_3$
(C) $Pb(NO_3)_2$ and $Cu(NO_3)_2$ (D) $NaNO_3$ and $Ca(NO_3)_2$

4. KO_2 finds use in oxygen cylinders used for space and submarines. The fact(s) related to such use of KO_2 is/are:
(A) it produces O_2 (B) It produces O_3
(C) It absorbs CO_2 (D) It absorbs both CO and CO_2

5. Which of the following are correct?
(A) 
(B) 
(C) 
(D) 

6. Which of the following statements are correct about stability of chelates–
(A) As the number of rings in complex increases, stability of chelate also increases
(B) A chelate having five membered ring is more stable if it contains double bonds.
(C) A chelate having six membered ring is more stable than if it does not contain double bonds.
(D) Chelating ligands are at least bidentate ligands.

7. Which of the following is/are extracted by electrolytic reduction?
(A) Cu (B) Al
(C) Mg (D) Ag

Space for rough work

8. Choose the incorrect statement:
   (A) Salicylic acid (o–Hydroxybenzoic acid) is much stronger acid than its m-,p-isomers and benzoic acid itself.
   (B) Acidity of salicylic acid is due to steric inhibition of resonance, as – OH group forces – COOH out of the plane of ring
   (C) The orbitals which are in the same plane take part in resonance
   (D) All the resonating structures have real existence

9. Rotation around the bond (between the underlined atoms) is restricted in:
   (A) $\underline{C}_2H_4$ (B) $H_2\underline{O}_2$
   (C) $\underline{C}_2H_2$ (D) $\underline{C}_2H_6$

10. Select the coloured compound(s):
   (A) $K_2Cr_2O_7$ (B) $CuSO_4.5H_2O$
   (C) $\left[V\left(H_2O\right)_6\right]Cl_3$ (D) $\left[Cu\left(NH_3\right)_4\right]SO_4$

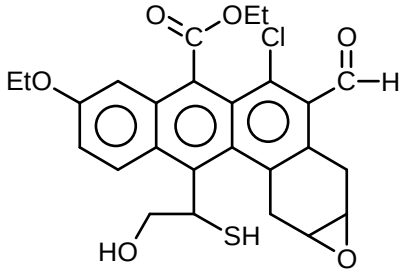
SECTION – C

(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Find the total number of species having planar shape.
   $I_3^+, XeF_4, SF_4, C_2F_4, H_2O_2, BrF_5, SO_3, NOCl, ClF_3, F_2CO, XeF_5^-$

2. How many moles of Grignard reagent are consumed in given compound?



Space for rough work

3. How many monochloro compounds including stereoisomers will give propene on treatment with Mg/ether followed by H_2O ?

4. Find the sum of optically active isomers of both $\left[Pt(gly)_2Cl_2\right]$ and $\left[Co(en)_2Cl_2\right]$.

5. A polyhydric alcohol is acylated by using ethanoyl chloride in presence of alkali. It was observed that molecular mass of alcohol is increased to 237% on acylation. If the molecular mass of polyhydric alcohol is 92, then the number of –OH groups present in polyhydric alcohol are____

6. For the synthesis of ammonia at 300 K :

$N_2(g) + 3H_{2(g)} \longrightarrow 2NH_3(g)$

Calculate the value of $\Delta G°$ in Kcal and give your answer in magnitude by using the following data:

	N_2	H_2	NH_3
ΔH_f° (Kcal/mole)	0	0	-10
$\Delta S°$ (Cal/K-mole)	40	30	45

7. Calculate the change in pressure (in atm) when 2 mole of NO and 16 g O_2 in a 6.25 litre originally at 27°C react to produce the maximum quantity of NO_2 possible according to the equation.

$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$ (Take R = $\frac{1}{12}$ ltr. Atm/mol K)

8. Electrons in a sample of H – atoms make transitions from state n = x, to some lower excited state. The emission spectrum from the sample is found to contain only the lines belonging to a particular series. If one of the photons had an energy of 0.6375 eV. Then find the value of x.

[Take 0.6375 eV = $\frac{3}{4}$ × 0.85eV]

9. If the atomic number of the inert gas is XY in which the total number of d-electrons is equal to the difference in the number of total p- and s- electrons. Find |X-Y|.

10. For the reaction $3A(g) \rightarrow 2B(g) + 2C(g)$, starting with pure A, pressure after 9 min was 387.5 mm of Hg and after a long time was 400mm of Hg. Calculate half life of A in min.

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

1. If $2f(x)+xf\left(\frac{1}{x}\right)-2f\left(\left|\sqrt{2}\sin\pi\left(x+\frac{1}{4}\right)\right|\right)=4\cos^2\left(\frac{\pi x}{2}\right)+x\cos\frac{\pi}{x},\forall x\in R=\{0\}$, then which of the following statement(s) is/are true?

(A) $f(2)+f\left(\frac{1}{2}\right)=1$ (B) $f(2)+f(1)=0$

(C) $f(2)+f(1)=f\left(\frac{1}{2}\right)$ (D) $f(1)f\left(\frac{1}{2}\right)f(2)=1$

2. Let ABC be a non obtuse triangle such that AB > AC & $\angle B=45^o$. Let O and I denote the circumcenter and incentre of ΔABC respectively. Suppose that $\sqrt{2}OI=AB-AC$, then the possible values of sin $\angle BAC$ is/are

(A) $1/\sqrt{2}$ (B) $\sqrt{\frac{4\sqrt{2}-2}{2}}$

(C) $\frac{\sqrt{4\sqrt{2}-2}}{2}$ (D) 1/2

3. $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ intersects the coordinate axis at points A, B and C resp. If ΔPQR has midpoints A, B, & C then

(A) centroid of ΔABC & ΔPQR coincide

(B) foot of normal to ΔABC from O is circumcentre of ΔPQR

(C) area$(\Delta PQR)=2\sqrt{a^2b^2+b^2c^2+c^2a^2}$

(D) incentres of ΔABC and ΔPQR coincide

Space for rough work

4. $\bar{x}\times\bar{y}=\bar{a},\ \bar{y}\times\bar{z}=\bar{b},\ \bar{x}.\bar{b}=\gamma,\bar{x}\ .\ \bar{y}=1\&\ \bar{y}.\bar{z}=1$ then

(A) $\bar{y}=\dfrac{\bar{a}\times\bar{b}}{\gamma}$ (B) $\bar{x}=\dfrac{1}{y^2}\left[\bar{y}-\bar{a}\times\bar{y}\right]$

(C) $\bar{z}=\dfrac{1}{y^2}\left[\bar{y}+\bar{b}\times\bar{y}\right]$ (D) $\bar{x}=\dfrac{1}{\gamma^2}\left[\bar{y}+\bar{a}\times\bar{y}\right]$

5. If the value of the definite integral $\int_0^1 {}^{207}C_7x^{200}(1-x)^7\,dx$ is equal to $\dfrac{1}{k}$ where $k\in N$, then k is divisible by

(A) 13 (B) 16

(C) 9 (D) 52

6. A function $f:R\to R$ satisfies the equation $f(x+y)=f(x)+f(y)\forall x,y\in R$ & is continuous through out the domain. If $I_1+I_2+I_3+I_4+I_5=450$ where $I_n=n\int_0^n f(x)dx$, then

(A) $f(x)=4x$ (B) $f(x)=2x$

(C) $f'(0)=4$ (D) $f'(0)=2$

7. $A(\bar{a}),B(\bar{b}),C(\bar{c})$ are position vector of vertices of ΔABC such that $\bar{a}.\bar{b}+\bar{a}.\bar{c}=\overline{a.a}+\bar{b}.\bar{c}$. Also $|\bar{b}-\bar{c}|+|\bar{a}-\bar{b}|=2|\bar{a}-\bar{c}|$. Inradius of triangle is 1 unit, then which of the following statement is/are correct?

(A) $\sin^2B+\sin^2C=1$ where B = $|\underline{ABC}$ & C = $|\underline{ACB}$

(B) Diameter of circumcircle of ΔABC is 5 units

(C) Area of ΔABC is 6 sq. units

(D) $\dfrac{\text{Area of }\Delta ABC}{\text{semi perimeter of }\Delta ABC}=2$

Space for rough work

8. The curves $y = ax^3 + 4x (a \neq 0)$ & $xy = 1$ touch each other at points P and Q, where P is in first quadrant. A triangle is formed by drawing a tangent to $y = ax^3 + 4x$ at point Q and coordinate axes. Then which of the following is/are correct?
(A) The value of a is – 4
(B) The area of the triangle formed is 2
(C) The value of a is – 2
(D) The area of the triangle formed is 1

9. If the value of definite integral $\int_{-1}^{1} \cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)\left(\cot^{-1}\frac{x}{\sqrt{1-\left(x^2\right)^{|x|}}}\right)dx = \pi^2\left(\frac{\sqrt{a}-\sqrt{b}}{c}\right)$,
where a, b, c, $\in$ N in their lowest form then
(A) a = 1 (B) b = 2
(C) a = 2 (D) b = 1

10. Let $A = \begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}$ & $(A+I)^{50} - 50A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then
(A) a + d =2 (B) a + b = 1
(C) b + c = 0 (D) a+ d = 1

SECTION – C

(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. $\frac{\tan 1}{\cos 2} + \frac{\tan 2}{\cos 4} + ... + \frac{\tan 1024}{\cos 2048} = \tan\lambda - \tan\mu$, then $\left[\frac{\lambda-\mu}{1000}\right] = ...$, [ ]G.I.F .

2. A triangle ABC has positive integer sides, $\angle A = 2\angle B$ and $\angle C > 90^o$, then the minimum length of the perimeter of ABC is 70 + K then K is equal to

Space for rough work

3. The value of $\sum_{k=0}^{7}\left(\frac{{}^{7}C_k}{{}^{14}C_k}\sum_{r=k}^{14}\left({}^{r}C_k\right)\left({}^{14}C_r\right)\right)$ is $(k)^{k+1}$. Then k is

4. If the value of $\int_0^1 \frac{\sin^{-1}\sqrt{x}}{x^2-x+1}dx$ is $\frac{\pi^2}{\sqrt{n}}$ $(\text{where } n \in N)$, then the value of $\frac{n}{27}$ is

5. The sum of the following n term series for $\theta = 30^o$

$\cos^2\theta+\cos^2\theta\cos(2\theta)+\cos^3\theta(\cos 3\theta)+.....+\cos^n\theta\cos(n\theta)$ is $(n+1)\left(\sin\frac{n\pi}{6}\right)\left(\frac{1}{k}\right)^n$, then k is

6. A tourist takes a trip through a city in stages each stage consist of three segments of length 100 meters separated by right turns of 60°. Between the last segment of one stage and first segment of the next stage, the tourist makes a left turn of 60°. The distance the tourist will be from his initial position after 2017 stages is 100 k. Then k is

7. Let $A_n (n \in N)$ be a matrix of order $(2n-1)\times(2n-1)$ such that $a_{ij} = 0\ \forall i \neq j$ & $a_{ij} = n^2+i+1-2n\ \forall i = j$, where a_{ij} denotes the element of ith row & j^{th} column of A_n. Let T_n = $(-1)^n \times$ (sum of all the element of A_n). Then the value of $\left[\frac{\sum_{n=1}^{102} T_n}{520200}\right]$, where $[.]$ represents the G.I.F, is

8. The minimum area bounded by the function $y = f(x)$ and $y = kx+9$, $(k-R)$, where f satisfies the relation $f(x+y) = 2f(x)+xf(y)+y\sqrt{f(x)}\ \forall x,y \in R$ & $f'(0) = 0$ is 9 A, value of A is

9. Let $f(x)$ be a differentiable function such that $f'(x)$ is symmetrical about the line 2x = 5 and f(1) = 2, f(4) = 6 then $\int_2^3 f(x)dx$ is equal to

10. If the radius of circumcircle of the ΔTPQ where PQ is the chord of contact corresponding to point T with respect to circle $x^2+y^2-2x+4y-11=0$ is b units. Then the minimum distance of T from director circle of the given circle is $12-4\sqrt{x}$, then k is equal to

Space for rough work